



Climate Crusade

SENG 401 - Software Architecture

Developers:

Stephan Agwai , Aadil Bashir, Arol Nokam Wafo

Gauthier Appaix, Ebitimi Dambo,

Emmanuel Idoro

Table of Contents

Requirements Artifact	3
General Scenario	3
Level Specific Scenarios	6
Level 1: Acid Downpour	6
Level 2: Deep Purge	8
Level 3: Sunken Wells	10
Level 4: Space Junk	12
Design Artifact	14
Testing Artifact	18
Requirements Traceability Matrix	18
Test Cases - Scripted Tests (General)	21
Test Cases - Scripted Tests (Level)	27
Bug Reports	43
Resolution	48

Generative AI Usage Statement

Our team used generative AI tools responsibly to support the development of *Climate Crusade*. These tools were used as collaborative aids to enhance creativity, productivity, and learning.

ChatGPT (OpenAI) was used to generate image prompts for cutscenes, platforms, sprites, objects, and backgrounds. It also assisted in refining documentation, improving code comments, and polishing written content.

ElevenLabs was used to generate voiceovers for cutscenes and background music during gameplay, enhancing immersion and user experience.

Canva AI and NVIDIA Nemo were used to animate static images for cutscenes and instructional sequences, improving visual storytelling.

The use of these tools enabled rapid prototyping, supported creative design, and allowed our team to focus on system architecture and gameplay development. Challenges included refining AI-generated assets for consistency and verifying outputs to ensure accuracy and originality. Overall, generative AI enhanced our learning experience while maintaining academic integrity.

Game Link: <https://saagee-studios.github.io/climate-crusade/>

GitHub Link: <https://github.com/SAAGEE-Studios/climate-crusade>

Requirements Artifact

General Scenario

User Stories	
ID	User Story
Accounts & Access 🗝️	
US-1	As a player, I want to log in to my account so that I can access the game and my saved progress.
US-2	As a new player, I want to create an account so that I can access the game and have my progress saved.
US-3	As a player, I want to delete my account so that I can permanently remove my data.
US-4	As a player, I want to logout of my account so that I can switch users or exit securely.
US-5	As a first time player, I want an introductory cutscene to play automatically so that I understand the game's premise.
Level Selection & Progress 🗺️	
US-6	As a player, I want to select and start unlocked so that I can progress through the game.
US-7	As a player, I want to view my level progress on the level selection screen so that I can see which levels are completed and which remain.
Progression & Persistence ★	
US-8	As a player, I want to collect stars within a level so that I can increase my overall progress.
US-9	As a player, I want my collected stars to be saved automatically when I complete a level so that my progress persists between sessions.
US-10	As a player, I want my earned stars to contribute to the Earth's health so that my actions visibly impact the game world.
Experience & Feedback 🎬	
US-11	As a player, I want short cutscenes to play at key moments so that the story progresses alongside gameplay.
US-12	As a player, I want to receive clear feedback upon completing a level so that I understand my performance and impact.

US-13	As a player, I want to pause the game so that I can choose to resume or exit the level.
US-14	As a player, I want to return to the level select screen from the title card so that I can change levels.

Functional Requirements		
ID	Functional Requirement	US. ID
Accounts & Access 🗝️		
FR-1	The system shall require users to log in with a valid username and password before accessing the game.	US-1
FR-2	The system shall allow new users to create an account.	US-2
FR-3	The system shall allow users to log out and return to the login screen.	US-3
FR-4	The system shall allow users to permanently delete their account and saved data.	US-4
FR-5	The system shall play an introductory cutscene for first-time users.	US-5
Level Selection & Progress 🗺️		
FR-6	The system shall allow players to select and start unlocked levels from the level selection screen.	US-6
FR-7	The system shall display each player's level progress on the level selection screen.	US-7
Progression & Persistence ★		
FR-8	The system shall allow players to collect stars during gameplay.	US-8
FR-9	The system shall save the number of stars collected when a level is completed.	US-9
FR-10	The system shall update Earth's health based on the total number of stars earned.	US-10
Experience & Feedback 🎬		
FR-11	The system shall play short cutscenes at key moments in the game.	US-11
FR-12	The system shall display feedback to the player after completing a level.	US-12

FR-13	The system shall allow the player to press the Esc key during gameplay to pause the level and display options to resume or exit.	US-13
FR-14	The system shall allow the player to press the Esc key on a level title card to return to the Level Select screen.	US-14

Non-Functional Requirements		
ID	NF Req.	Description
NF-1	Security	The system shall require password verification before allowing account deletion.
NF-2	Usability	The system shall provide clear feedback messages for login, signup, and account deletion actions.
NF-3	Performance	The system shall load and transition between scenes without noticeable delay under normal operating conditions.
NF-4	Reliability	The system shall preserve the highest number of stars earned for a level and prevent overwriting with lower values.
NF-5	Accessibility	Cutscenes shall include subtitles to support players with hearing impairments.
NF-6	Maintainability	The system shall separate game logic, user interface logic, and backend communication to improve maintainability.
NF-7	Inclusivity	The system shall incorporate culturally relevant settings and language within levels to promote global environmental awareness.

Level Specific Scenarios

Level 1: Acid Downpour



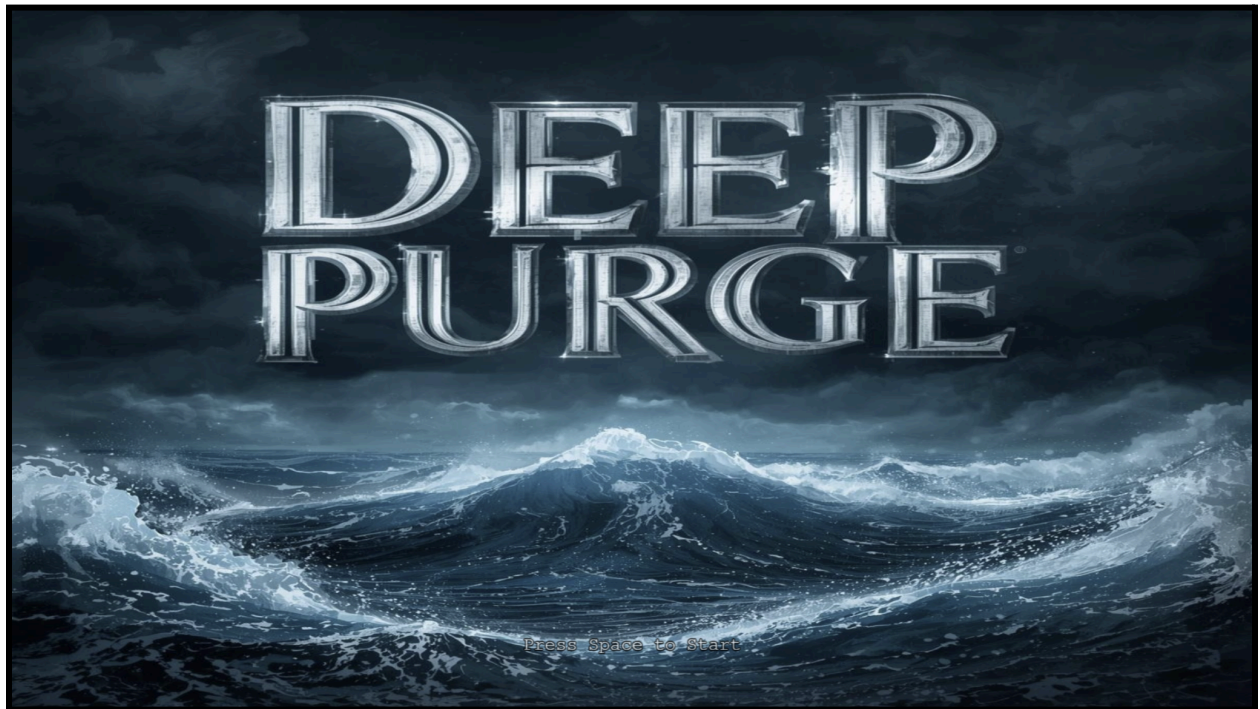
User Stories	
ID	User Story
Accounts & Access 	
US-1	As a player, I want to move my character left, right, and jump so that I can navigate the platforms safely.
US-2	As a player, I want to collect stars so that I can track my progression and score.
US-3	As a player, I want to activate buttons so that I can unlock platforms and solve level puzzles.
US-4	As a player, I want to toggle levers so that I can turn emission systems on/off.
US-5	As a player, I want to see my remaining lives during gameplay so that I can manage my survival.
US-6	As a player, I want to see a pause menu so that I can pause the game and choose to exit or resume.

US-7	As a player, I want to be educated about ways I can contribute to acid rain prevention on the end screen.
------	---

Functional Requirements		
ID	Functional Requirement	US. ID
FR-1	The player can move left/right and jump, with collisions applied to platforms and boundaries.	US-1
FR-2	Stars appear at designated positions and can be collected by the player.	US-2
FR-3	Buttons can be activated when the player is near them and presses the interaction key, unlocking or enabling associated platforms.	US-3
FR-4	Levers can be toggled when the player is near them and presses the interaction key, switching emission systems on/off.	US-4
FR-5	The game displays the number of remaining lives and updates dynamically when the player takes damage.	US-5
FR-6	The pause menu can be opened during gameplay, allowing the player to resume or exit the level.	US-6
FR-7	Save progress asynchronously to the database for stars collected in the level.	US-2
FR-8	At the end of the level, display a set of educational tips about how the player can contribute to reducing acid rain, in a clearly readable format.	US-7

Non-Functional Requirements		
ID	NF Req.	Description
NF-1	Responsiveness	Player input (movement, jump, lever/button activation) must respond immediately without delay.
NF-3	Usability	Controls, interactions, and HUD elements should be intuitive so players can focus on gameplay rather than figuring out mechanics.
NF-4	Cultural Inclusivity	Cutscenes, backgrounds, and dialogue reflect the Thai setting to provide cultural representation and context.
NF-5	Visual Accessibility	All UI elements (hearts, stars, buttons) should be clearly visible and distinguishable for players of varying visual acuity.

Level 2: Deep Purge



User Stories	
ID	User Story
Core Gameplay	
US-1	As a player, I want the fishing hook to swing continuously side-to-side so that I have to time my casts carefully.
US-2	As a player, I want to cast the hook using the SPACE bar or screen tap so that I can reach items deep in the water.
Level Constraints & UI	
US-3	As a player, I want to see a countdown timer on the screen so that I know how much time I have left to clean the ocean.
Win/Lose Conditions	
US-4	As a player, I want to clear the level by successfully hooking 3 stars so that I can progress in the game.

Functional Requirements		
ID	Functional Requirement	US. ID
FR-1	The system shall continuously oscillate the hook's angle while in the idle state.	US-1
FR-2	The system shall extend the hook downwards along its current vector when the SPACE bar or pointer down event is triggered.	US-2
Level Constraints & UI		
FR-3	The system shall display a timer that counts down from 120 seconds, updating the HUD every second.	US-3
Win/Lose Conditions		
FR-4	The system shall monitor the number of collected stars and trigger the win state overlay once the count reaches 3.	US-4

Non-Functional Requirements		
ID	NF Req.	Description
NF-1	Usability	The system shall display a temporary on-screen hint ("SPACE or TAP to cast hook") that automatically fades out after 3 seconds to guide new players without permanently cluttering the UI.
NF-2	Performance	The dynamic rendering of the rope (calculated using 24 geometric segments) shall maintain a smooth framerate during extension and retraction sequences.
NF-3	Reliability	The collision detection system shall consistently and accurately register a catch when the hook's world coordinates are within a 28-pixel radius of a star.
NF-4	Aesthetics	The game shall provide dynamic visual feedback, such as floating particle effects and scaling animations on the collected stars, upon successfully completing the level.

Level 3: Sunken Wells



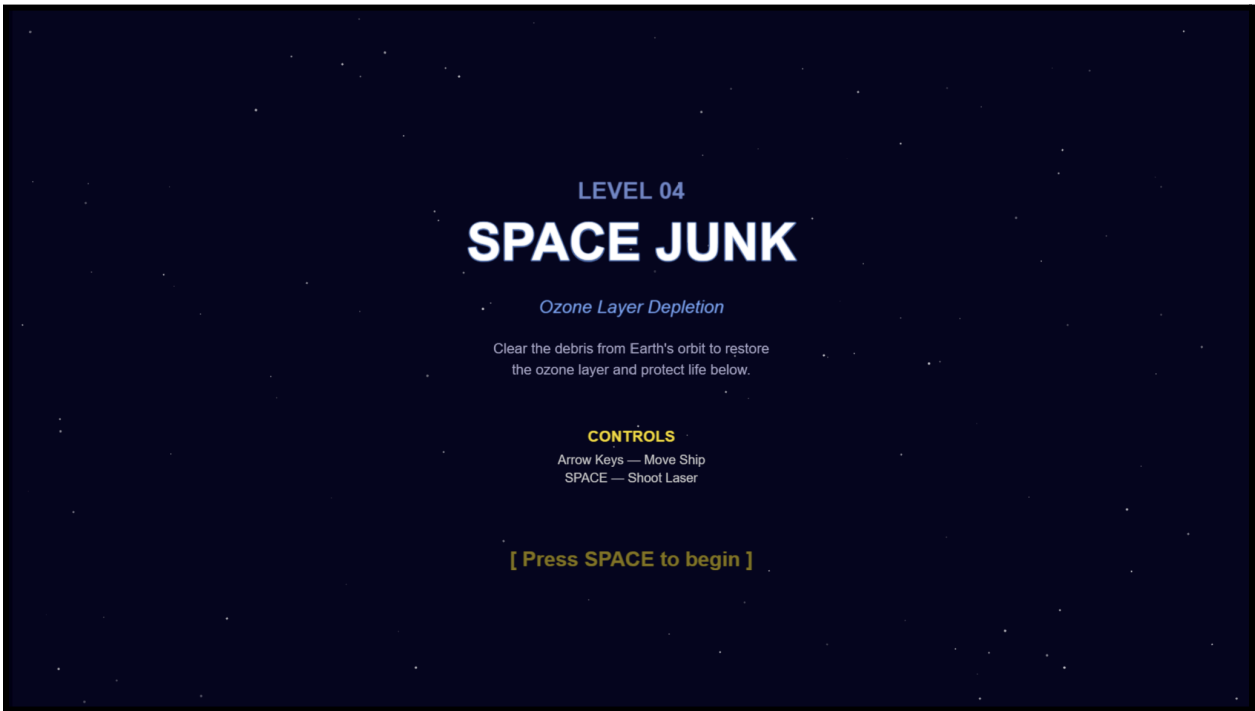
User Stories	
ID	User Story
Accounts & Access 	
US-1	As a player, I want to be able to move up, down, left and right.
US-2	As a player, I want to be able to dig through the blocks.
US-3	As a player, I want to see how much oxygen I have left,so that I know when I have failed..
US-4	As a player, I want to know which powerups are active.
US-5	As a player, I want to collect stars to progress the Earth's health.
US-6	As a player, I want to reach the water at the bottom of the level to win so that I have a clear goal to work toward.
US-7	As a player, I want environmental hazards to challenge me during my descent so that the level remains engaging.
US-8	As a player, I want my collected stars to be saved when I complete the level so that my progress is preserved.

US-9	As a player, I want to see a pause menu so that I can pause the game and choose to exit or resume.
------	--

Functional Requirements		
ID	Functional Requirement	US. ID
FR-1	The player is affected by gravity, can jump, and move left and right	US-1
FR-2	The player can dig up blocks.	US-2
FR-3	The UI displays the active powerups.	US-4
FR-4	The UI displays the remaining oxygen	US-3
FR-5	The player can collect stars and the UI displays them	US-5
FR-6	When the player's oxygen reaches zero, the level shall end in a loss state and display a game-over overlay.	US-3
FR-7	When the player reaches the bottom of the grid, the level shall end in a win state and display a win overlay.	US-6
FR-8	The level shall periodically trigger a row-based wind hazard that warns the player before dealing damage if they occupy the affected row.	US-7
FR-9	The system shall save the number of stars collected only when the level is completed in a win state.	US-8
FR-10	The player shall be able to open a pause menu during gameplay to resume or exit the level.	US-10

Non-Functional Requirements		
ID	NF Req.	Description
NF-1	Performance	Player input (movement, jump, lever/button activation) must respond immediately without delay.
NF-2	Accessibility	The gameplay should be simple and easy to pick up.
NF-3	Accessibility	The interface should use clear visual cues and readable text.
NF-4	Reliability	Scene transitions shall occur without crashes or errors.

Level 4: Space Junk



User Stories	
ID	User Story
Accounts & Access 	
US-1	As a player, I want to move my ship left, right, and up/down so that I can navigate the starfield and avoid debris.
US-2	As a player, I want to shoot lasers so that I can destroy space debris.
US-3	As a player, I want to collect stars so that I can track my progression and score.
US-4	As a player, I want to see my current score so that I can track my performance during the level.
US-5	As a player, I want to see my health so that I can monitor my ship's survivability.
US-6	As a player, I want to see which wave I am on so that I can anticipate increasing difficulty.
US-7	As a player, I want to view educational facts between waves so that I can learn about space debris and ozone protection.
US-8	As a player, I want a pause menu so that I can pause the game and choose to exit

	or resume.
--	------------

Functional Requirements		
ID	Functional Requirement	US. ID
FR-1	The ship can move in all four directions within screen boundaries.	US-1
FR-2	The ship can shoot laser bullets with a fire rate limit.	US-2
FR-3	Debris of varying sizes (small, medium, large) spawn according to wave configurations.	US-2
FR-4	Collectible stars spawn randomly once per wave.	US-3
FR-5	Score increases when debris is destroyed; display updates dynamically.	US-4
FR-6	Ship health decreases when debris reaches the bottom of the screen; display updates dynamically.	US-5
FR-7	Waves are announced visually with subtitles describing debris difficulty.	US-6
FR-8	Educational fact overlay appears after each wave and pauses gameplay until dismissed.	US-7
FR-9	Players can pause the game, resume, or exit via keyboard controls.	US-8
FR-10	At the end of the level, display total score, debris cleared, and stars collected in a completion screen.	US-3, US-4
FR-11	Save progress asynchronously to the database for stars collected in the level.	US-3

Non-Functional Requirements		
ID	NF Req.	Description
NF-1	Usability	Controls, interactions, and HUD elements should be intuitive for players to quickly understand objectives and mechanics.
NF-2	Visual Accessibility	All HUD elements (health, score, stars, waves) must be clearly visible and distinguishable.

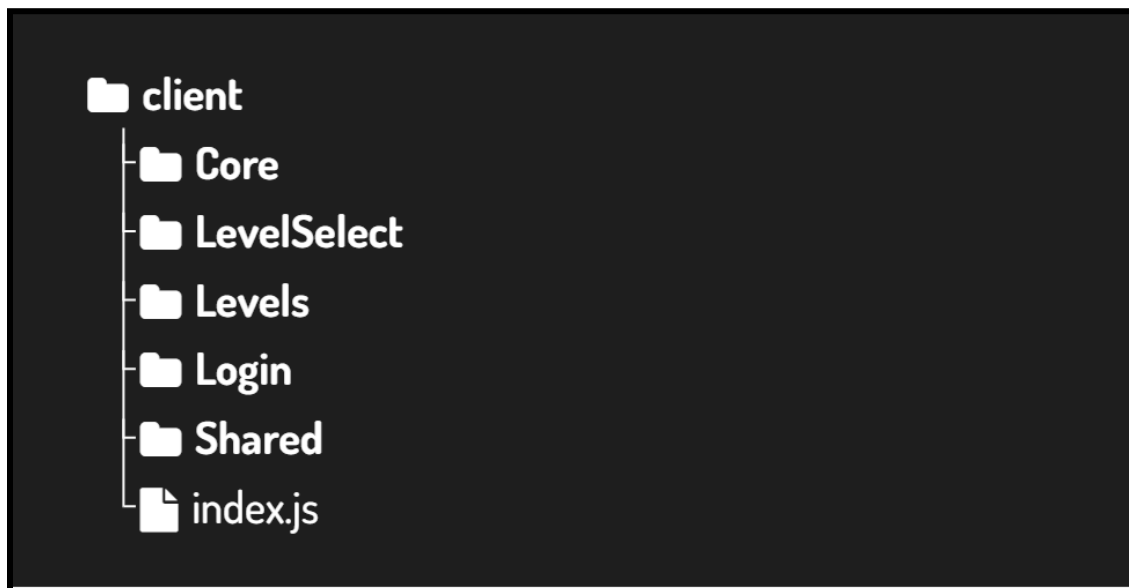
Design Artifact

Climate Crusade uses a three-tier client–server architecture consisting of a presentation layer (client), an application layer (server), and a data layer (database). The database is contained within the server directory for convenience but operates as a separate persistence layer. It is accessed exclusively by the application layer.

The presentation layer is responsible for rendering the game and handling user interaction. It is structured into carefully organized folders that separate responsibilities across the front-end. The Core module coordinates transitions between major parts of the game, including login, level selection, and gameplay. It manages scene transitions, level discovery, and configuration loading.

The Login module handles user authentication and session initialization and serves as the first point of interaction when the game starts. The Level Selection module allows players to navigate between available levels and displays progress data retrieved from the server. The Levels directory contains all playable game levels. Each level is implemented as a self-contained module with its own scenes, assets, and configuration file. This modular structure allows levels to remain independent of one another and supports scalability.

The Shared directory contains reusable assets and utilities used across multiple modules, including global UI components, audio files, and shared visual assets. The presentation layer is implemented using JavaScript, Phaser, HTML, and CSS.



The backend is implemented using Node.js with the Express framework. The main server file (index.js) defines REST API endpoints that handle communication between the client and the database. These endpoints are responsible for authentication, progress storage, account management, and data retrieval.

The data layer uses SQLite to store user accounts and level progress. SQL queries are used to insert, update, retrieve, and validate data. The database is accessed only through the server to ensure proper validation, security, and data integrity.

Program flow operates as follows:

The client sends an HTTP request to the server through a defined API endpoint. The server processes the request, performs any necessary validation or logic, and interacts with the database if required. The database returns the requested data to the server, which then sends a structured response back to the client. The client uses this data to update the user interface or game state.

This architecture cleanly separates presentation, application logic, and data management, improving maintainability, scalability, and security.

Justification:

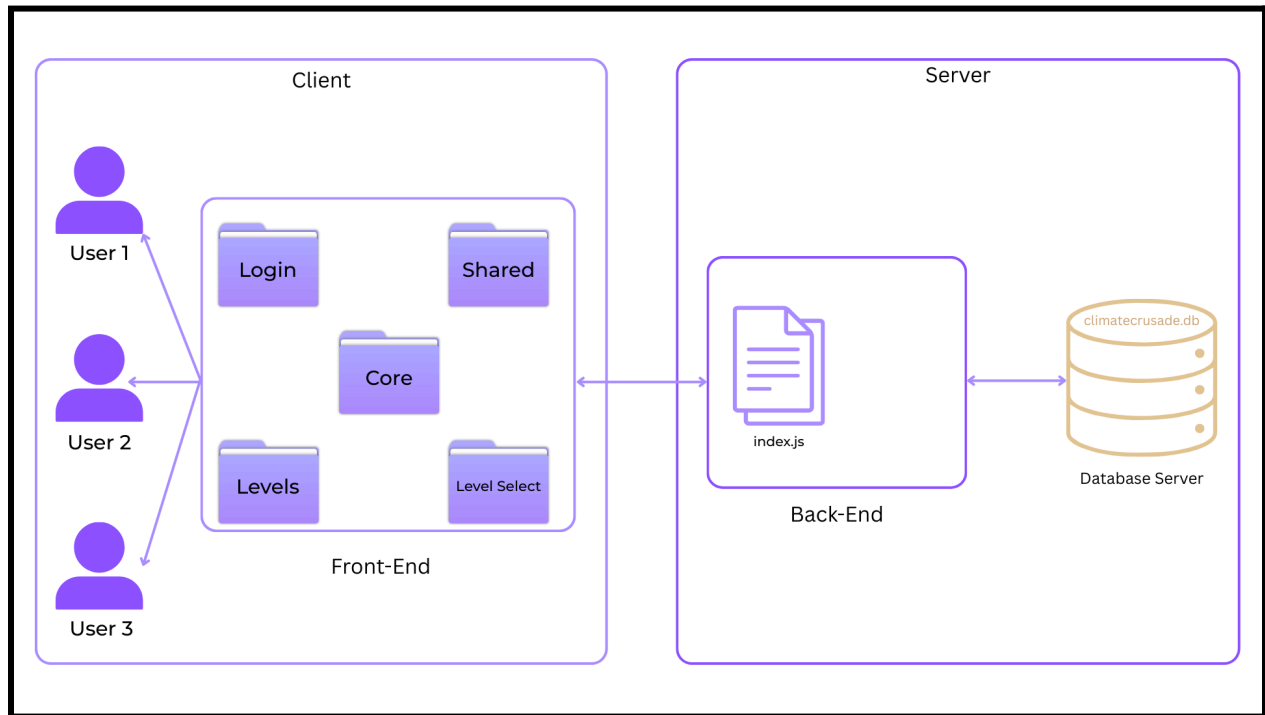
From the beginning, Climate Crusade was designed as a web-based application inspired by interactive games we had played growing up. To support this goal, we chose a three-tier client-server architecture.

This architectural style separates the system into three distinct layers: the presentation layer (client), the application layer (server), and the data layer (database). The presentation layer handles rendering, user interaction, and gameplay logic using Phaser. The application layer, implemented using Node.js and Express, manages authentication, validation, and API communication. The data layer, implemented using SQLite, is responsible for persistent storage of user accounts and level progress. Although the SQLite database file resides within the server project directory for deployment simplicity, it remains logically separated as the dedicated data layer within the three-tier architecture.

Client-server architecture was chosen because it provides a clear separation of concerns. Sensitive operations such as password hashing and data validation are handled exclusively on the server, improving security. At the same time, the client focuses on rendering and user experience, ensuring smooth gameplay and responsive user experience.

This architecture also supports scalability and maintainability. New levels can be added without modifying authentication or database logic, and backend changes can be made independently of front-end design. By isolating data management from gameplay logic, the system remains modular and easier to extend.

System Architecture Diagram



This diagram illustrates the three-tier client-server architecture used in Climate Crusade.

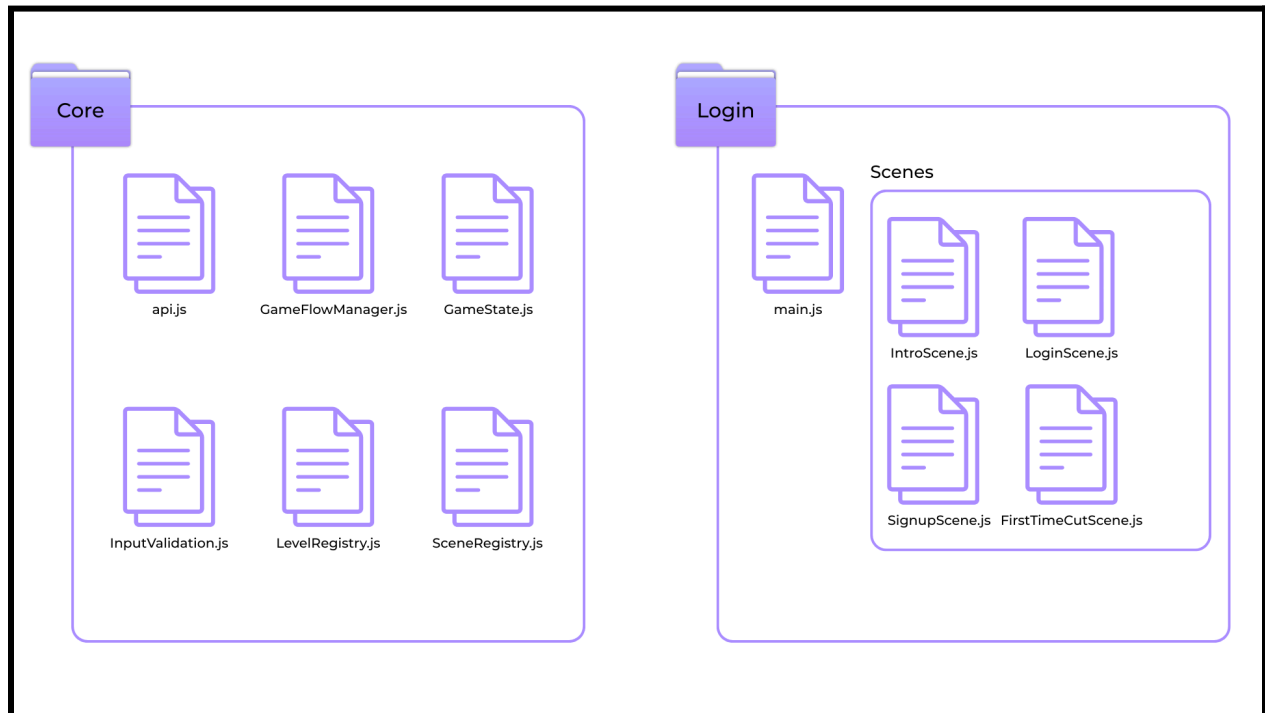
On the left is the Client (Presentation Layer), where users interact with the game. The front-end is built using Phaser and is divided into modules such as Login, Levels, Shared, Core, and Level Select. This layer handles rendering, user input, and gameplay logic.

On the right is the Server (Application Layer), implemented using Node.js and Express. The server manages authentication, validation, and communication between the client and the database.

The Database Server (Data Layer) stores persistent data, including user accounts and level progress (climatecrusade.db). The client does not directly access the database; all requests pass through the server.

This separation ensures security, modularity, and maintainability while supporting smooth gameplay and persistent user progress.

Front-End Structural Overview (In-depth look at the client)



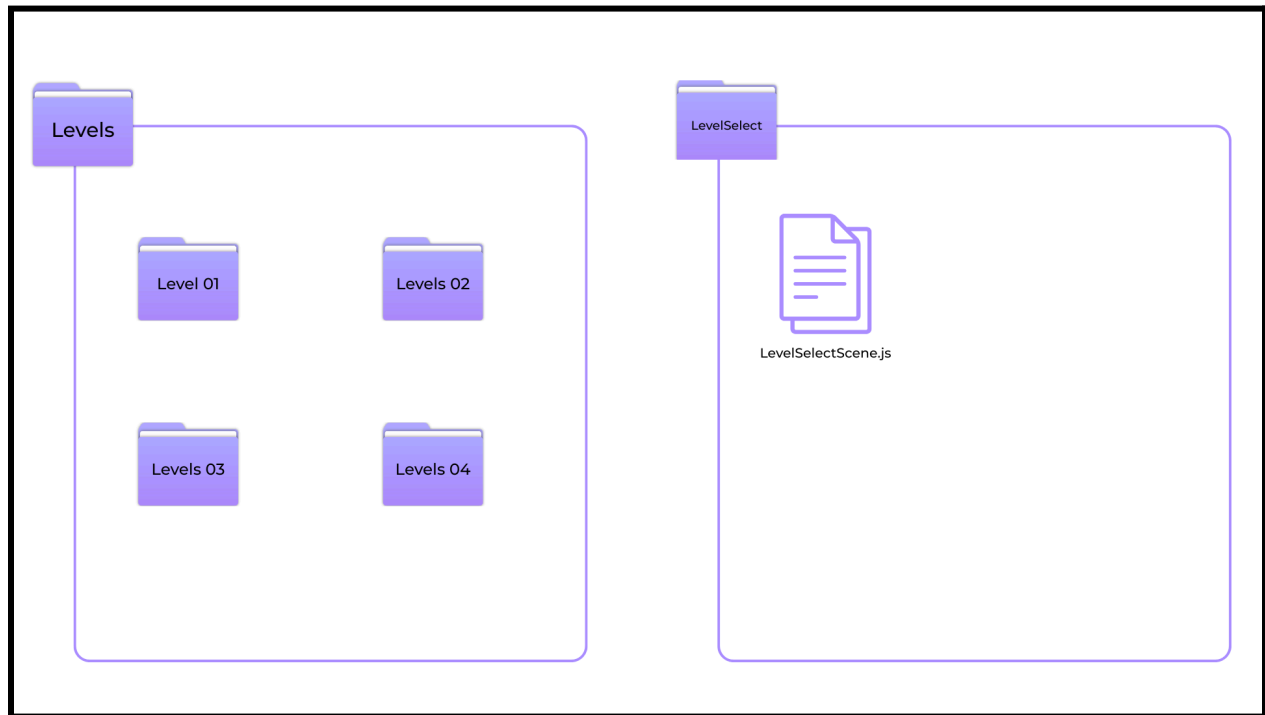
This diagram highlights the two most important front-end modules in Climate Crusade: Core and Login.

The Core module contains shared system logic that supports the entire game. It manages API communication (`api.js`), global state (`GameState.js`), scene transitions (`GameFlowManager.js`), input validation, and scene/level registration. This module acts as the foundation of the front-end architecture.

The Login module represents the system entry point. It contains the initialization file (`main.js`) and the primary entry scenes, including `IntroScene`, `LoginScene`, `SignupScene`, and `FirstTimeCutScene`. These scenes handle user authentication and onboarding before transitioning to gameplay.

Although not all front-end components are shown, this diagram focuses on the architectural core and entry flow of the system. Together, these modules demonstrate how shared logic is separated from scene-specific functionality, improving modularity and maintainability.

Levels and Level Selection Structure (In-depth look at the client)



This diagram highlights the gameplay structure within the front-end of Climate Crusade.

The Levels module contains the individual game levels (Level 01–Level 04). Each level is self-contained, with its own assets, mechanics, and environmental theme. This modular design allows new levels to be added without modifying existing ones, supporting scalability and maintainability.

The LevelSelect module contains `LevelSelectScene.js`, which acts as the central hub for choosing levels. It connects the authentication flow to the gameplay experience and manages navigation between different levels.

By separating level selection from individual level implementations, the system maintains clear organization and separation of concerns. The Level Select scene handles navigation logic, while each level focuses only on its gameplay mechanics and objectives.

Testing Artifact

Requirements Traceability Matrix

Req. ID	Requirement Description	Design Reference	Development Reference	Test Case ID	Status
FR1	The system shall require users to log in with a valid username and password before accessing the game.	Presentation Layer, Login Module	Login screen	TC001 TC002 TC003	C... ▾
FR2	The system shall allow new users to create an account.	Presentation Layer, Account Creation	Signup Screen	TC004 TC005 TC006	C... ▾
FR3	The system shall allow users to log out and return to the login screen.	Core Module, Login Module	Logout Button, Login Screen,	TC007	C... ▾
FR4	The system shall allow users to permanently delete their account and saved data.	Account Management, Database Module	Delete Account Button, Database	TC008	C... ▾
FR5	The system shall play an introductory cutscene for first-time users.	Core Module, Presentation Layer	Introductory Cutscene Code, First Time User Logic	TC009 TC010	C... ▾
FR6	The system shall allow players to select and start unlocked levels from the level selection screen.	Core Module, Level Selection Module	Level Selection Screen, Start Level	TC011	C... ▾
FR7	The system shall display each player's level progress on the level selection screen.	Core Module, Level Selection Module, Database Module	Progress Displaying Logic, Database	TC012	C... ▾
FR8	The system shall allow players to collect stars during gameplay.	Level Module, Database Module	Star Collection Logic, Database	TC013	C... ▾
FR9	The system shall save the number of stars collected when a level is completed.	Core Module, Database Module	Save Progress Logic	TC014 TC015	C... ▾
FR10	The system shall update Earth's health based on the total number of stars earned.	Core Module, Database Module	Earth Health Logic, Star Impact Logic	TC016	C... ▾
FR11	The system shall play short cutscenes at key moments in the game.	Core Module, Level Module, Presentation Layer	Cutscene Screen	TC017	C... ▾
FR12	The system shall display feedback to the player after completing a level.	Presentation Layer, Gameplay Module	Level Complete Screen	TC018	C... ▾

FR-13	The system shall allow the player to press the Esc key during gameplay to pause the level and display options to resume or exit.	Gameplay Module	Level Gameplay Screen	TC019	C... ▾
FR-14	The system shall allow the player to press the Esc key on a level title card to return to the Level Select screen.	Gameplay Module	Level Title Card	TC020	C... ▾

Level 1 Requirements Traceability Matrix			
ID	Functional Requirement	TC. ID	Status
FR-1	The player can move left/right and jump, with collisions applied to platforms and boundaries.	TC-L1-001	Completed ▾
FR-2	Stars appear at designated positions and can be collected by the player.	TC-L1-006, TC-L1-003	Completed ▾
FR-3	Buttons can be activated when the player is near them and presses the interaction key, unlocking or enabling associated platforms.	TC-L1-007	Completed ▾
FR-4	Levers can be toggled when the player is near them and presses the interaction key, switching emission systems on/off.	TC-L1-008, TC-L1-009	Completed ▾
FR-5	The game displays the number of remaining lives and updates dynamically when the player takes damage.	TC-L1-002, TC-L1-004	Completed ▾
FR-6	The pause menu can be opened during gameplay, allowing the player to resume or exit the level.	TC-L1-011	Completed ▾
FR-7	Save progress asynchronously to the database for stars collected in the level.	TC-L1-005, TC-L1-010	Completed ▾
FR-8	At the end of the level, display a set of educational tips about how the player can contribute to reducing acid rain, in a clearly readable format.	TC-L1-012	Completed ▾

Level 2 Requirements Traceability Matrix			
ID	Functional Requirement	TC. ID	Status
FR-1	The system shall continuously oscillate the hook's angle while in the idle state.	TC-L2-001	Completed ▾
FR-2	The system shall extend the hook downwards along its current vector when the SPACE bar or pointer down event is triggered	TC-L2-002, TC-L2-003	Completed ▾
FR-3	The system shall display a timer that counts down from 120 seconds,	TC-L2-004	Completed ▾

	updating the HUD every second.		
FR-4	The system shall monitor the number of collected stars and trigger the win state overlay once all trash is picked up.	TC-L2-006	Completed ▾

Level 3 Requirements Traceability Matrix			
ID	Functional Requirement	TC. ID	Status
FR-1	The player is affected by gravity, can jump, and move left and right	TC-L3-001	Completed ▾
FR-2	The player can dig up blocks.	TC-L3-002	Completed ▾
FR-3	The UI displays the active powerups.	TC-L3-007, TC-L3-008	Completed ▾
FR-4	The UI displays the remaining oxygen	TC-L3-003, TC-L3-004, TC-L3-005	Completed ▾
FR-5	The player can collect stars and the UI displays them	TC-L3-006	Completed ▾
FR-6	When the player's oxygen reaches zero, the level shall end in a loss state and display a game-over overlay.	TC-L3-011	Completed ▾
FR-7	When the player reaches the bottom of the grid, the level shall end in a win state and display a win overlay.	TC-L3-012	Completed ▾
FR-8	The level shall periodically trigger a row-based wind hazard that warns the player before dealing damage if they occupy the affected row.	TC-L3-010, TC-L3-009	Completed ▾
FR-9	The system shall save the number of stars collected only when the level is completed in a win state.	TC-L3-013, TC-L3-014	Completed ▾
FR-10	The player shall be able to open a pause menu during gameplay to resume or exit the level.	TC-L3-015	Completed ▾

Level 4 Requirements Traceability Matrix			
ID	Functional Requirement	TC. ID	Status
FR-1	The ship can move in all four directions within screen boundaries.	TC-L4-001	Completed ▾
FR-2	The ship can shoot laser bullets with a fire rate limit.	TC-L4-002	Completed ▾
FR-3	Debris of varying sizes (small, medium, large) spawn according to wave configurations.	TC-L4-003	Completed ▾

FR-4	Collectible stars spawn randomly once per wave.	TC-L4-004	Completed ▾
FR-5	Score increases when debris is destroyed; display updates dynamically.	TC-L4-005	Completed ▾
FR-6	Ship health decreases when debris reaches the bottom of the screen; display updates dynamically.	TC-L4-006	Completed ▾
FR-7	Waves are announced visually with subtitles describing debris difficulty.	TC-L4-007	Completed ▾
FR-8	Educational fact overlay appears after each wave and pauses gameplay until dismissed.	TC-L4-008	Completed ▾
FR-9	Players can pause the game, resume, or exit via keyboard controls.	TC-L4-009	Completed ▾
FR-10	At the end of the level, display total score, debris cleared, and stars collected in a completion screen.	TC-L4-010	Completed ▾
FR-11	Save progress asynchronously to the database for stars collected in the level.	TC-L4-011	Completed ▾

Test Cases - Scripted Tests (*General*)

TC001	Valid Login✔	
Preconditions	User account already exists and user is on the login screen.	
Procedure	Action	Expected Result
1	Enter a valid username and valid password.	Input is shown for username but dots are shown for password.
2	Click the login button.	User is logged in and enters the game.
Expected Result	User accessed the game level selector.	
Post Condition	User accessed the game level selector.	
TC002	Invalid Login, Wrong Username✔	
Preconditions	User is on the login screen.	
Procedure	Action	Expected Result
1	Enter an incorrect username and a valid password.	Input is shown for username but dots are shown for password.
2	Click the login button.	Login is rejected and an error message is shown.
Expected Result	User is not allowed into the game with an invalid username.	
Post Condition	User is not allowed into the game with an invalid username.	
TC003	Invalid Login, Wrong Password✔	
Preconditions	User is on the login screen and the account exists.	
Procedure	Action	Expected Result
1	Enter a valid username and an incorrect password.	Input is shown for username but dots are shown for password.
2	Click the login button.	Login is rejected and an error message is shown.
Expected Result	User is not allowed into the game with an invalid password and remains on the login screen.	
Post Condition	User is not allowed into the game with an invalid password and remains on the login screen.	
TC004	Successful Account Creation✔	

Preconditions	User is on the signup screen.	
Procedure	Action	Expected Result
1	Enter a new valid username and password.	Input is accepted.
2	Submit the form.	New account is created successfully and user gets thrown onto the login page.
Expected Result	User account is created and can be used on the login page to sign in and remains on the login screen.	
Post Condition	User account is created and can be used on the login page to sign in and remains on the login screen.	

TC005	Failed Account Creation due to invalid credentials 	
Preconditions	User is on the signup screen.	
Procedure	Action	Expected Result
1	Enter invalid account information.	Input attempt is accepted.
2	Submit the form.	Account creation is rejected and an error message is shown.
Expected Result	System prevents account creation, user remains on the signup screen.	
Post Condition	System prevents account creation, user remains on the signup screen.	

TC006	Failed account creation due to duplicate credentials 	
Preconditions	User is on the signup screen.	
Procedure	Action	Expected Result
1	Enter duplicate account information.	Input attempt is accepted.
2	Submit the form.	Account creation is rejected and an error message is shown.
Expected Result	System prevents duplicate account creation, user remains on the signup screen.	
Post Condition	System prevents duplicate account creation, user remains on the signup screen.	

TC007	Successful Logout 	
Preconditions	User is logged in.	
Procedure	Action	Expected Result
1	Open the menu or level select screen.	Current game screen is visible.

	2	Click the logout button.	Logout action is triggered.
Expected Result		User is logged out and returned to the login screen.	
Post Condition		User is logged out and returned to the login screen.	
TC008		Successful Account Deletion 	
Preconditions		User is on the level selection screen.	
Procedure	Action		Expected Result
	1	Select delete account and confirm deletion with password input.	Account and saved data are permanently removed and user is thrown into login screen.
	2	Attempt to log in again with the deleted account.	Login fails because the account no longer exists.
Expected Result		User account and related data are deleted successfully.	
Post Condition		User account and related data are deleted successfully.	
TC009		First Time Intro Plays 	
Preconditions		User is logging in with a first-time account.	
Procedure	Action		Expected Result
	1	Log in with a first-time user account.	System identifies the account as first-time.
	2	User needs to wait a moment as the video loads.	Introductory cutscene plays.
	3	Wait for the cutscene to finish or skip.	User proceeds into the game selection screen.
Expected Result		Players with new accounts see the introductory cutscene	
Post Condition		Players with new accounts see the introductory cutscene	
TC010		Returning User Skips Intro 	
Preconditions		User is logging in with a returning account.	
Procedure	Action		Expected Result
	1	Log in with a returning user account.	User is sent to level selection screen without intro cutscene.
Expected Result		Returning users skip the intro cutscene and displays the level selection screen.	
Post Condition		Returning users skip the intro cutscene and displays the level selection screen.	

TC011	Start Unlocked Level 	
Preconditions	User is logged in and at least one level is unlocked.	
Procedure	Action	Expected Result
1	Open the level selection screen.	Available levels are displayed.
2	Select an unlocked level.	Selection is accepted.
3	Start the level.	Chosen level loads successfully.
Expected Result	Player can play unlocked levels.	
Post Condition	Player can play unlocked levels.	
TC012	Level Progress Displays 	
Preconditions	User is on level selection screen and has completed at least one level	
Procedure	Action	Expected Result
1	Check the displayed progress.	Displayed progress matches saved progress.
2	Compare completed and incomplete levels.	Level progress display is accurate.
Expected Result	Player progress is shown correctly on the level selection screen.	
Post Condition	Player progress is shown correctly on the level selection screen.	
TC013	Stars Collected During Gameplay 	
Preconditions	Player is inside a level that contains stars.	
Procedure	Action	Expected Result
1	Move the player into a star collectible.	Player reaches the star.
2	Collect the star.	Star is collected and counted.
3	Check the in-level count or indicator.	Star count updates correctly.
Expected Result	Stars can be collected during gameplay.	
Post Condition	Stars can be collected during gameplay.	
TC014	Stars Saved After Level Completion 	
Preconditions	Player can complete a level after collecting stars.	
Procedure	Action	Expected Result
1	Collect one or more stars.	Stars are counted during gameplay.
2	Complete the level.	Collected stars are saved.

3	Return to level select or reload progress.	Saved star count remains recorded.
Expected Result	Stars earned in a completed level are saved.	
Post Condition	Stars earned in a completed level are saved.	
TC015	Incomplete Level Progress Not Saved ✓	
Preconditions	Player starts a level but does not finish it.	
Procedure	Action	Expected Result
1	Collect one or more stars.	Stars are counted during the session.
2	Exit or fail before completing the level.	Incomplete run ends.
3	Reload the level or saved progress.	Incomplete run is not saved as completed progress.
Expected Result	Incomplete level progress is not saved as completed progress.	
Post Condition	Incomplete level progress is not saved as completed progress.	
TC016	Earth Health Heals When Collecting Stars ✓	
Preconditions	Player has a profile where Earth health can be viewed and stars can be earned.	
Procedure	Action	Expected Result
1	Collect stars during gameplay or through completed levels.	Total earned stars increases.
2	Check the Earth health indicator.	Earth health value is displayed.
3	Compare Earth health before and after.	Earth health increases based on stars earned.
Expected Result	Earth health updates positively when stars are collected.	
Post Condition	Earth health updates positively when stars are collected.	
TC017	Cutscenes Trigger at Appropriate Times ✓	
Preconditions	Player is able to reach a key story moment in the game.	
Procedure	Action	Expected Result
1	Progress through gameplay.	Player approaches a key event.
2	Reach the key moment.	Cutscene trigger condition is met.
Expected Result	Cutscenes appear at the intended key moments.	
Post Condition	Cutscenes appear at the intended key moments.	

TC018	Post Level Feedback Displayed 	
Preconditions	Player can complete a level.	
Procedure	Action	Expected Result
1	Complete the level.	Level completion flow is triggered.
2	Review the displayed information.	Player can see performance or progress details.
Expected Result	Player sees feedback after finishing a level.	
Post Condition	Player sees feedback after finishing a level.	
TC019	Player Exits Level During Gameplay 	
Preconditions	Player can exit a level during gameplay by pressing escape	
Procedure	Action	Expected Result
1	Start the level	Level opens up and player can begin.
2	Press 'esc'	Player is prompted to either 'exit' or 'continue'
Expected Result	Player can exit the level or continue playing	
Post Condition	Player goes to title card if 'Y' is selected, and continues playing if 'N' is selected.	
TC020	Player Exits On Title Card 	
Preconditions	Player can exit the title card screen by pressing escape	
Procedure	Action	Expected Result
1	Press Play on a level	Title Card is displayed
2	Press 'esc'	Player is taken back to the level select scene
Expected Result	Player sees the level select screen and can select another level	
Post Condition	Player is on the level select screen	

Test Cases - Scripted Tests (Level)

TC-L1-001	Player Movement✔	
Preconditions	Player has started Level 1	
Procedure	Action	Expected Result
	1 Press left arrow key.	Player moves left.
	2 Press right arrow key.	Player moves right.
	3 Press up arrow key.	Player jumps upwards.
Expected Result	Movement and jump behaves correctly.	
Post Condition	Player can move around in the level.	
TC-L1-002	Acid Contact Reduce A Life✔	
Preconditions	Player has started Level 1 and has 3 lives	
Procedure	Action	Expected Result
	1 Observe heart UI.	3 hearts are visible.
	2 Walk into the acid.	Player turns red, then respawns.
	Observe heart UI.	One heart is darkened, 2 remains lit.
Expected Result	Touching acid reduces lives by 1 and respawns the player.	
Post Condition	Player has 2 lives remaining.	
TC-L1-003	Collected Stars Persist Through Respawn✔	
Preconditions	Player has started Level 1 and at least one star is reachable.	
Procedure	Action	Expected Result
	1 Collect one star during gameplay	Star disappears, count is tracked.
	2 Walk into the acid	Player respawn.
	3 Observe the star positions in the level.	Collected star doesn't reappear.
.Expected Result	Collected star persists after death.	
Post Condition	Star count is unchanged after respawn.	

TC-L1-004	Three Deaths Trigger Game Over✔	
Preconditions	Player has started level one with 3 lives	
Procedure	Action	Expected Result
	1 Touch acid once.	Respawned with 2 lives.
	2 Touch acid a 2nd time.	Respawned with 1 life.
	3 Touch acid a 3rd time.	Game over overlay appears.
	4 Observe overlay content.	"GAME OVER" title, star icons, "Stars Will Be Discarded" message, and "Press Esc to Exit" are shown.
Expected Result	Three acid deaths trigger the game over screen.	
Post Condition	Level is finished, player cannot move.	
TC-L1-005	Stars Not Saved on Game Over✔	
Preconditions	Player starts Level 1 with 0 stars previously saved.	
Procedure	Action	Expected Result
	1 Collect 1 or more stars.	Stars tracked in session.
	2 Die 3 times to trigger game over.	Game over overlay appears with "Stars Will Be Discarded".
	3 Press ESC to exit and check level select.	Level select shows 0 stars for Level 1.
Expected Result	Stars collected during a lost run are not saved.	
Post Condition	Saved progress is unchanged.	
TC-L1-006	Star Collection Updates Session Count✔	
Preconditions	Player is inside Level 1 and at least one star is visible.	
Procedure	Action	Expected Result
	1 Move player into a star.	Star disappears.
	2 Collect a second star.	Second star disappears.
	3 Collect a third star.	Third star disappears.
	4 Observe star icons in the end overlay upon winning.	All 3 star icons are shown as filled.
Expected Result	All 3 stars can be collected and are reflected in the win overlay.	

Post Condition	All 3 stars can be collected and are reflected in the win overlay.	
TC-L1-007	Button Activates Secret Platform✔	
Preconditions	Player is near a push button and the associated secret platform is hidden.	
Procedure	Action	Expected Result
	1 Stand close to the button and press SPACE.	Secret platform fades in and becomes solid.
	2 Walk or jump onto the now-visible platform.	Player lands on it as a normal platform.
	3 Return to the button and press SPACE again.	Platform fades out and becomes non-solid.
	4 Attempt to stand where the platform was.	Player falls through the now-invisible area.
Expected Result	Button toggles the secret platform on and off.	
Post Condition	Platform state matches the last button press.	
TC-L1-008	Both Levers Down Triggers Win✔	
Preconditions	Player has started Level 1 and both levers are in the up position.	
Procedure	Action	Expected Result
	1 Navigate to the left lever and press SPACE.	Left lever goes down. Level does not end.
	2 Navigate to the right lever and press SPACE.	Right lever goes down.
	3 Observe the screen immediately after.	Win overlay appears.
	4 Observe overlay content.	Star icons, lives remaining, climate tips, and "Press Esc to Exit" shown.
Expected Result	Level completes when both levers are simultaneously in the down position.	
Post Condition	Physics paused, player cannot move, progress is saved.	

TC-L1-009	Level State Persists Through Respawn✔	
Preconditions	Player has started Level 1 and at least one level can be reached.	
Procedure	Action	Expected Result
	1 Toggle the top left lever down.	Level is in the down position.
	2 Touch acid to trigger respawn.	Level restarts.
	3 Observe the lever after respawn.	Lever is still in the down position.
Expected Result	Lever states are preserved across deaths.	
Post Condition	Player can proceed to toggle the remaining levers to win.	
TC-L1-010	Stars Saved on Level Completion✔	
Preconditions	Player starts Level 1 with 0 stars previously saved	
Procedure	Action	Expected Result
	1 Collect 2 stars during the run.	Stars tracked in session.
	2 Toggle both levers down to complete the level.	Win overlay appears.
	3 Press ESC to exit and check the level select.	Level select shows 2 stars for Level1.
Expected Result	Stars collected during a winning run are saved to the backend.	
Post Condition	Saved progress reflects the completed run.	
TC-L1-011	Pause Menu Opens and Closes✔	
Preconditions	Player is inside Level 1 during active gameplay.	
Procedure	Action	Expected Result
	1 Press ESC during gameplay.	Pause menu appears with "Exit Level?", Y and N prompts.
	2 Observe that the game is paused.	Player doesn't move.
	3 Press N.	Pause menu closes, gameplay resumes.
	4 Press ESC again, then press Y.	Player is returned to the Level 1 entry scene
Expected Result	Pause menu correctly handles both resume and exit.	
Post Condition	Player either continues the run or exits cleanly.	

TC-L1-012			Climate Tips Display on Win Overlay✔
Preconditions			Player can complete the level by toggling both levers.
Procedure		Action	Expected Result
	1	Complete the level	Win overlay appears
	2	Observe the overlay	Three educational tips exist in the overlay.
Expected Result			All three climate tips are displayed and fade in after the win.
Post Condition			Tips are fully visible and readable
TC-L2-001			Hook Oscillates Continuously✔
Preconditions			Player has started Level 2 and the level is in the idle state.
Procedure		Action	Expected Result
	1	Observe the hook before casting.	Hook swings continuously side-to-side.
	2	Cast hook then continue observing after the hook returns.	Hook casts then returns then keeps oscillating.
Expected Result			The hook continuously oscillates.
Post Condition			The hook remains in oscillation until the player casts and continues oscillating after it returns.
TC-L2-002			Hook Cast Using SPACE Key✔
Preconditions			Player is inside Level 2 and the hook is oscillating in the idle state.
Procedure		Action	Expected Result
	1	Press the SPACE bar.	Hook begins extending downward along its current direction.
	2	Observe hook motion.	When hook returns it is in a state where it can be cast again and continues oscillating.
Expected Result			Pressing SPACE casts the hook along its current direction.
Post Condition			The hook completes a cast and continues normal gameplay behavior.
TC-L2-003			Hook Cast Using Mouse Input✔

Preconditions	Player is inside Level 2 on a device/browser setup that supports pointer input.	
Procedure	Action	Expected Result
1	Mouse left click input	Hook begins extending downward along its current direction.
2	Observe the cast.	When hook returns it is in a state where it can be cast again and continues oscillating.
Expected Result	Pressing left mouse button casts the hook along its current direction.	
Post Condition	The hook completes a cast and continues normal gameplay behavior.	
TC-L2-004	Timer Displays and Counts Down✔	
Preconditions	Player has entered Level 2.	
Procedure	Action	Expected Result
1	Observe the timer value.	Timer begins at 120 seconds and counts down once per second.
Expected Result	The timer is displayed and counts down correctly from 120 seconds.	
Post Condition	Timer continues decreasing until level end or timeout.	
TC-L2-005	Player Can Hook Game Objects✔	
Preconditions	Player is inside Level 2.	
Procedure	Action	Expected Result
1	Cast the hook toward the object.	Hook extends toward the object.
2	Hook hits object and retrieves it.	Object is successfully hooked and dragged back to the boat.
3	Observe the star count or score indicator.	Star collection count increases correctly.
Expected Result	The player can successfully hook and retrieve a collectible object.	
Post Condition	The collected object is counted toward level progress.	
TC-L2-006	Level Ends When All Stars Are Collected✔	
Preconditions	Player is inside Level 2 and only the required stars remain to complete the level.	
Procedure	Action	Expected Result

1	Successfully hook and collect the first star.	Star collection count increases to 1.
2	Successfully hook and collect the second star.	Star collection count increases to 2.
3	Successfully hook and collect the third star.	Star collection count increases to 3.
4	Observe the screen after the third collection.	Win state overlay is triggered.
Expected Result		The level enters the win state when all stars are collected.
Post Condition		The level enters the win state when all stars are collected.
TC-L2-007		Game Instructions Appear on Start✔
Preconditions		Player starts Level 2 from the beginning.
Procedure	Action	Expected Result
1	Start Level 2.	Level loads and instructions appear.
2	Click anywhere on the screen or press space.	Game instructions disappear.
Expected Result		After clearing instructions gameplay starts.
Post Condition		Gameplay starts.
TC-L4-001		Ship Movement Within Screen Boundaries✔
Preconditions		Player has started Level 4.
Procedure	Action	Expected Result
1	Move the ship left, right, up, and down.	Ship responds correctly in all four directions.
2	Move the ship toward each screen edge.	Ship approaches the screen boundaries.
3	Attempt to move past the screen edges.	Ship does not move outside the visible play area.
Expected Result		The ship can move in all four directions and stays on screen.
Post Condition		The ship can move in all four directions and stays on screen.

TC-L4-002			Ship Shoots Lasers With Fire Rate Limit✔
Preconditions			Player is inside Level 4 and gameplay is active.
Procedure	Action	Expected Result	
	1 Press the shoot control once.	A laser bullet is fired from the ship.	
	2 Rapidly press the shoot control multiple times.	Lasers fire only as allowed by the fire rate limit.	
	3 Continue firing over time.	Shooting remains consistent without exceeding the intended rate.	
Expected Result			The ship fires lasers correctly and respects the fire rate limit.
Post Condition			Player can continue firing lasers normally.
TC-L4-003			Debris Spawns According to Wave✔
Preconditions			Player has started Level 4 and a wave begins.
Procedure	Action	Expected Result	
	1 Start the level and observe the first wave.	Debris begins appearing on screen.	
	2 Continue through wave progression.	Debris continues spawning until wave ends	
	3 Compare later waves to earlier waves.	Later waves contain more difficult debris patterns or increased challenge.	
Expected Result			Debris spawns according to wave setup, including different debris sizes.
Post Condition			Waves continue spawning as gameplay progresses.
TC-L4-004			Collectible Star Spawns Once Per Wave✔
Preconditions			Player is inside Level 4 and at least one wave is active.
Procedure	Action	Expected Result	
	1 Observe the level during an active wave.	A collectible star appears during the wave.	
	2 Continue observing the same wave after the star appears.	No second star appears in that same wave.	
	3 Advance to the next wave.	A new star may spawn for the new wave.	
Expected Result			Exactly one collectible star spawns per wave.

Post Condition	Star spawns and is collectible each level.	
TC-L4-005	Score Updates When Debris Is Destroyed✔	
Preconditions	Player is inside Level 4 and debris is present.	
Procedure	Action	Expected Result
	1 Destroy a piece of debris using lasers.	Debris is removed when destroyed.
	2 Observe the score display.	Score increases.
	3 Destroy additional debris.	Score continues updating.
Expected Result	Destroying debris increases the score and updates correctly.	
Post Condition	Score reflects total debris destroyed so far.	
TC-L4-006	Health Decreases When Debris Reaches Bottom✔	
Preconditions	Player is inside Level 4 with visible health.	
Procedure	Action	Expected Result
	1 Allow debris to fall without destroying it.	Debris continues moving downward.
	2 Let debris reach the bottom of the screen.	Ship health decreases.
	3 Observe the health display.	Ship health updates whenever damage is received.
Expected Result	Health decreases whenever debris reaches the bottom of the screen.	
Post Condition	Player continues the level with reduced health.	
TC-L4-007	Wave Announcement Appears With Subtitle✔	
Preconditions	Player has started Level 4 and is progressing between waves.	
Procedure	Action	Expected Result
	1 Start a new wave.	A wave announcement appears on screen.
	2 Observe the text shown with the announcement.	Subtitle describes the wave.
	3 Wait for gameplay to continue.	Announcement disappears after a few moments.
Expected Result	Each wave is visually announced with descriptive subtitle text.	

Post Condition	Gameplay proceeds into the announced wave.	
TC-L4-008	Educational Fact Overlay Appears After Each Wave✔	
Preconditions	Player is able to complete at least one wave in Level 4.	
Procedure	Action	Expected Result
	1 Complete a wave.	Educational fact overlay appears.
	2 Click to continue.	Gameplay resumes.
Expected Result	An educational fact overlay appears after each wave and pauses gameplay until dismissed.	
Post Condition	Player continues into the next wave after dismissal.	
TC-L4-009	Exit Prompt Opens, Resumes, and Exits✔	
Preconditions	Player is inside Level 4 during active gameplay.	
Procedure	Action	Expected Result
	1 Press escape key.	Exit level prompt pops up.
	2 Press N.	Exit level prompt closes and gameplay continues.
	3 Press escape key.	Exit level prompt pops up.
	4 Press Y.	Player leaves the level and returns to the level select screen.
Expected Result	The exit prompt allows the player to continue or leave the level.	
Post Condition	Player either resumes gameplay or exits successfully.	
TC-L4-010	Completion Screen Displays Final Results✔	
Preconditions	Player can finish Level 4.	
Procedure	Action	Expected Result
	1 Progress through the level until completion conditions are met.	Level completion is triggered.
	2 Observe the completion screen.	Completion screen appears.
	3 Review the displayed results.	Total score, debris cleared, and stars collected are shown.
Expected Result	The completion screen displays the player’s total score, debris cleared, and stars collected.	
Post Condition	Player sees a summary of Level 4 performance.	

TC-L4-011	Stars Saved After Level Completion✔	
Preconditions	Player is able to collect stars and complete Level 4.	
Procedure	Action	Expected Result
	1 Collect one or more stars during Level 4.	Stars are counted during gameplay.
	2 Complete the level successfully.	Level completion flow is triggered.
Expected Result	Stars collected in Level 4 are saved after successful completion.	
Post Condition	Saved progress reflects the completed run.	
TC-L3-001	Player Movement - Left, Right, Up✔	
Preconditions	Player has started Level3 and the spawn area is visible.	
Procedure	Action	Expected Result
	1 Press left arrow or A.	Player moves left.
	2 Press right arrow or D.	Player moves right
	3 Press up arrow or W while standing on a surface.	Player jumps upwards.
Expected Result	Player moves left, right and up correctly.	
Post Condition	Player remains in bounds and responds to directional input	
TC-L3-002	Digging - Remove Block in Dig Direction✔	
Preconditions	Player has entered the grid and is adjacent to a block tile.	
Procedure	Action	Expected Result
	1 Hold a directional key towards a block and press E.	The block in that direction is removed.
	2 Wait for the dig cooldown (500ms) and press E again.	A second block is removed.
	3 Press E twice in quick succession.	Only one block is removed.
Expected Result	Player can dig in the held direction, subject to cooldown.	
Post Condition	Dug cells are empty and the player can move through them	
TC-L3-003	Digging - Trap Tile Deals Damage✔	
Preconditions	Player is adjacent to a trap tile and over 40 oxygen.	
Procedure	Action	Expected Result

	1	Observe current oxygen level.	Oxygen is above 40.
	2	Hold direction toward the trap tile and press E.	Trap tile is removed.
	3	Observe oxygen after the dig.	Oxygen has decreased by the trap damage amount (10).
Expected Result		Digging a trap tile removes it and reduces oxygen.	
Post Condition		Player continues with reduced oxygen, trap tile is gone.	
TC-L3-004		Oxygen Drain Activates Upon Entering the Grid ✓	
Preconditions		Player has started Level 3 and is in the spawn area above the grid.	
Procedure	Action		Expected Result
	1	Player has started Level 3 and is in the spawn area above the grid.	Oxygen does not drain.
	2	Player enters the grid.	Oxygen begins draining.
	3	Observe oxygen bar over several seconds.	Oxygen decreases continuously at a rate of 2 units per second.
Expected Result		Oxygen drain only activates once the player crosses into the grid	
Post Condition		Oxygen continues draining for the rest of the run	
TC-L3-005		Air Pickup Refills Oxygen ✓	
Preconditions		Player is in the grid with oxygen below maximum and an air pickup is present.	
Procedure	Action		Expected Result
	1	Note current oxygen level.	Oxygen is below 100.
	2	Move player into the air pickup.	Pickup is collected and disappears.
	3	Observe oxygen level.	Oxygen increases by approximately 33, capped at 100.
Expected Result		Air pickup refills oxygen and is consumed on collection.	
Post Condition		Pickup is gone, oxygen is higher.	
TC-L3-006		Star Collection Updates UI ✓	
Preconditions		Player is in the grid, with no stars collected and a star pickup is visible.	
Procedure	Action		Expected Result

	1	Observe star icons in the UI, they should be dim	Three dim star icons are displayed.
	2	Move the player into a star pickup.	Star is collected and disappears
	3	Observe the UI	One star icon is now fully lit
	4	Collect the remaining two stars.	All three star icons are fully lit.
Expected Result		Star icons in the HUD update to reflect collected stars.	
Post Condition		Stars are tracked and displayed correctly.	
TC-L3-007		Shield Powerup Blocks Damage✔	
Preconditions		Player has collected a shield pickup	
Procedure		Action	Expected Result
	1	Observe the shield icon	Icon is fully lit and shows 2 charges.
	2	Dig into a trap tile.	Trap damage is absorbed, oxygen does not decrease.
	3	Observe shield charge count.	Count decreases by 1.
	4	Dig a second trap tile.	Second hit is absorbed.
	5	Observe shield icon after second hit.	Shield deactivates, icon dims.
Expected Result		Shield absorbs up to 2 damage hits before deactivating.	
Post Condition		Shield is inactive, player takes normal damage from further hazards.	
TC-L3-008		Drill Powerup Reduces Dig Cooldown✔	
Preconditions		Player has collected a drill pickup.	
Procedure		Action	Expected Result
	1	Observe the drill icon	Icon is lit and labeled ON.
	2	Rapidly press E while adjacent to blocks.	Blocks are removed faster than normal
	3	Wait for the drill to expire (10 secs)	Icon dims and dig speed returns to normal
Expected Result		Drill reduces dig cooldown and expires after 10 seconds	
Post Condition		Dig cooldown returns to 500ms	
TC-L3-009		Wind Hazard Warning And Damage✔	

Preconditions	Player is inside the grid and has oxygen.	
Procedure	Action	Expected Result
1	Wait for a wind event to trigger	A red pulsing warning band appears on a row near the player.
2	Move player onto the warned row after the warning ends and the sandslide appears.	Player takes damage and oxygen decreases.
3	Wait for the sandslide visual to fade.	Asset fades out cleanly
Expected Result	Wind hazard shows a warning, damages the player once if on the active row, then fades.	
Post Condition	Row is no longer active, no further damage from this event.	
TC-L3-010	Wind Hazard and No Damage✔	
Preconditions	Player is inside the grid and has oxygen	
Procedure	Action	Expected Result
1	Observe the warning band row.	Warning appears on a specific row.
2	Ensure player is not on the warned row when the sandslide appears.	Player is on a different row during the damage window
3	Observe oxygen.	Oxygen does not decrease due to damage.
Expected Result	Player takes no damage when not occupying the active wind row.	
Post Condition	No oxygen loss, gameplay continues normally.	
TC-L3-011	Oxygen Depleted is Lose Condition✔	
Preconditions	Player is in the grid with low oxygen	
Procedure	Action	Expected Result
1	Allow oxygen to drain to zero without collecting air	Oxygen bar empties completely
2	Observe the screen.	End overlay appears with "OUT OF OXYGEN" title.
3	Observe the overlay content.	Stars collected, climate tips, and "Press Esc to Exit" are displayed.
Expected Result	Stars collected, climate tips, and "Press Esc to Exit" are displayed.	
Post Condition	Player can exit to the entry scene via ESC.	

TC-L3-012			Reach Water is Win Condition✔
Preconditions			Player is near the bottom of the grid.
Procedure	Action	Expected Result	
	1 Continue digging toward the bottom row	Grid continues downward	
	2 Move or fall player past the last row into the water	Level win is triggered after a short delay	
	3 Observe the overlay	"WATER REACHED" title is shown in green	
	4 Observe stars and tips in overlay	Collected stars and climate action tips are displayed	
Expected Result			Reaching the water triggers the win overlay with correct content.
Post Condition			Player can exit to the entry scene via ESC.
TC-L3-013			Stars Not Saved on Loss✔
Preconditions			Player starts Level 3 with 0 stars previously saved
Procedure	Action	Expected Result	
	1 Collect 1 or 2 stars during the run.	Star UI updates to reflect collected stars.	
	2 Allow oxygen to deplete to trigger the lose condition.	Lose overlay appears	
	3 Press ESC to return to the entry scene and navigate to level select.	Level select displays 0 stars for Level 3.	
Expected Result			Stars collected during a lost run are not saved to the backend.
Post Condition			Saved progress is unchanged from before the run
TC-L3-014			Stars Saved on Win✔
Preconditions			Player starts Level 3 with 0 stars previously saved.
Procedure	Action	Expected Result	
	1 Collect at least 1 star during the run.	Star UI updates.	
	2 Reach the water to trigger the win condition.	Win overlay appears.	
	3 Press ESC to exit and navigate to level select.	Level select displays the correct number of stars earned.	

Expected Result	Stars collected during a winning run are saved and reflected on the level select screen.	
Post Condition	Saved progress reflects the completed run.	
TC-L3-015	Pause Menu Opens and Closes Mid-Dive✔	
Preconditions	Player has dug several rows into the grid.	
Procedure	Action	Expected Result
	1 Press ESC during gameplay.	Pause menu appears centered on screen
	2 Observe that the game is paused	Player does not move, oxygen does not drain
	3 Press N to cancel	Pause menu closes and gameplay resumes
	4 Press ESC again and press Y to confirm exit	Player is returned to the Level 3 entry scene
Expected Result	Pause menu is always visible on screen and correctly handles resume and exit.	
Post Condition	Player either continues the run or exits cleanly.	

Exploratory Testing Charter

In addition to scripted testing, exploratory testing sessions were conducted to identify edge cases and usability issues. Findings were incorporated into refinements of gameplay mechanics and UI feedback.

Name	AroI NOKAM
Charter	<Explore level 1 gameplay> <in regular gameplay condition> <to discover if the functionality is as expected>
System setup	Windows 11 Chrome
Test start time	March 19th 2026, 7pm
Areas Tested	- Pause menu - Movement - Buttons and levers - Stars and saving functionality
How I tested	By playing the level normally
Defects	NA
Open questions	The stars are hard to miss, in a way that makes getting them trivial and unintentional. Placing them so the player would have to go out of their way to collect them could make the level more engaging.
Test stop time	March 19th 2026, 11pm
Other non-charter testing	Also successfully tested the cutscene and entryscene flow.

Name	Gauthier Appaix
Charter	<Explore level 2 gameplay> <in regular gameplay condition> <to discover if the functionality is as expected>
System setup	Windows 11 Chrome, Brave
Test start time	March 24th 2026, 4pm
Areas Tested	- Pause menu - Hook casting controls

	- Stars and saving functionality
How I tested	By playing the level normally
Defects	The hook asset appeared as a black box instead of displaying correctly. Pressing Esc exited the level directly instead of opening the pause menu first.
Open questions	The level is pretty short and everything is static. Could it be more engaging to have some movement to the objects to make it more challenging to catch the garbage and stars?
Test stop time	March 24th 2026, 6pm
Other non-charter testing	Also successfully tested the cutscene and entryscene flow.

Name	Arol NOKAM
Charter	<Explore level 3 gameplay> <in regular gameplay condition> <to discover if the functionality is as expected>
System setup	Windows 11 Chrome
Test start time	March 19th 2026, 7pm
Areas Tested	- Pause menu - Movement - Obstacles - Powerup mechanics - Stars and saving functionality
How I tested	By playing the level normally
Defects	The pause menu doesn't follow the camera, causing it to disappear when the player has dug past the initial rows. The stars are saved on loss instead of being discarded.
Open questions	Should the instructions card include arrow keys for movement since they're an alternative input? Could the player sprite look more detailed?
Test stop time	March 19th 2026, 11pm
Other non-charter testing	Also successfully tested the cutscene and entryscene flow.

Name	Gauthier Appaix
Charter	<Explore level 4 gameplay> <in regular gameplay condition> <to discover if the functionality is as expected>
System setup	Windows 11 Chrome, Brave
Test start time	March 23rd 2026, 4pm
Areas Tested	- Pause menu - Movement - Collision with space junk - Collision with screen boundaries - Space junk damage to earth - Stars and saving functionality
How I tested	By playing the level normally
Defects	NA
Open questions	Should the objects floating towards earth move faster depending on the level and the player be able to get upgrades to make it more engaging for the player?
Test stop time	March 23rd 2026, 6pm
Other non-charter testing	Also successfully tested the cutscene and entryscene flow.

Exploratory testing was conducted specifically for Levels 1–4, as gameplay mechanics are dynamic and involve real-time interactions, physics behavior, timing, and user decision-making. These aspects are less predictable and benefit from session-based exploration to uncover edge cases and usability issues.

The remaining system components, including authentication, API communication, and data persistence, were tested using structured scripted test cases with defined inputs, expected outputs, and validation steps. Since these components are deterministic, scripted testing provided sufficient coverage and verification.

For this reason, exploratory testing was focused on gameplay modules, where dynamic behavior makes exploratory approaches more effective.

Validation Discussion:

The test suite was created to check that the main features of Climate Crusade worked properly and matched the project requirements. It covered overall game features, such as login, account management, cutscenes, pause functionality, level progression, and movement. As well as covering level-specific mechanics, scoring, and win conditions.

Using scripted test cases helped keep testing organized and aligned with the functional requirements. The scripted test cases were also mapped to the functional requirements through the Requirements Traceability Matrix (RTM), ensuring that all implemented features were validated against the original system requirements.

Each test included preconditions, steps, test data, expected results, and postconditions, which made the tests clear and repeatable. Testing was evaluated by comparing the expected results with the actual results seen during execution. When the test passed, the feature being tested was considered validated for that case. When it did not, the issue was recorded as a bug and reviewed.

One challenge was that some bugs only appeared in specific gameplay situations, making them harder to catch. Structured tests made it clear how to test it and what to look for; however, when doing unscripted testing, it was unclear.

A limitation of the scripted tests is that they focus mainly on functional requirements, so they can't cover every possible edge case. This is why we made use of exploratory testing for unpredictable aspects of the game, like level gameplay.

Overall, the testing allowed us to prove that the game's main systems work and were tested in a clear, structured way, and that the results were used to improve the final product.

Bug Reports

Category	Label	Value
Bug ID	ID Number	BUG-001
	Name	Logout Button Disappears on Level Select Scene After Returning From a Level
	Reporter	Ebitimi Dambo
	Submit Date	March 12, 2026
Bug Overview	Summary	The Logout button is visible on the Level Select scene upon initial render. However, after entering a level and returning to the Level Select scene, the Logout button no longer appears. The issue persists until the page is refreshed.
Environment	Operating System	OS 26100.7840
	Browser	Chrome 145.0.7632.77
Bug Details	Steps to reproduce	1. Launch the application. 2. Log in successfully. 3. Navigate to the Level Select scene. 4. Confirm that the Logout button is visible. 5. Enter any level. 6. Exit the level and return to the Level Select scene. 7. Observe the UI.
	Expected results	The Logout button should consistently appear every time the Level Select scene is rendered.
	Actual results	The Logout button is missing after returning from a level. It only appears again after refreshing the browser.
Bug Tracking	Severity	Medium
	Priority	High
Notes	Notes	Likely caused by scene lifecycle handling or UI elements not being recreated in create() when the scene is re-entered. Possibly related to persistent state or object destruction during scene transitions.

Category	Label	Value
Bug ID	ID Number	BUG-002
	Name	Background Music Continues Playing After Leaving Login Scene
	Reporter	Ebitimi Dambo
	Submit Date	March 12, 2026
Bug Overview	Summary	Background music initiated in the Login scene continues playing even after navigating away from the Login scene into subsequent scenes.
Environment	Operating System	OS 26100.7840
	Browser	Chrome 145.0.7632.77
Bug Details	Steps to reproduce	1. Launch the application. 2. Observe background music playing on the Login scene. 3. Successfully log in. 4. Navigate to the Level Select scene or any level. 5. Listen for audio.
	Expected results	Background music from the Login scene should stop when transitioning to a different scene.
	Actual results	Login background music continues playing in subsequent scenes, resulting in overlapping or unintended audio.
Bug Tracking	Severity	Medium
	Priority	Medium
Notes	Notes	Likely caused by the audio object being global or not explicitly stopped during scene transition.

Category	Label	Value
Bug ID	ID Number	BUG-003
	Name	Intro cutscene loops until the user clicks skip.
	Reporter	Gauthier Appaix

	Submit Date	March 15, 2026
Bug Overview	Summary	The intro video loops on first login.
Environment	Operating System	Windows 11
	Browser	Brave
Bug Details	Steps to reproduce	1. Launch the application. 2. Make a new account 3. Log in with a new account to trigger cutscenes 4. Watch the full cutscene until it loops back to the beginning
	Expected results	Once the cutscene is done, it is expected to go to the level selection menu
	Actual results	Intro cutscene loops
Bug Tracking	Severity	low
	Priority	Medium
Notes	Notes	N/A

Category	Label	Value
Bug ID	ID Number	BUG-004
	Name	Level3 - Menu display
	Reporter	Arol Nokam
	Submit Date	March 19, 2026
Bug Overview	Summary	The display for the menu becomes dysfunctional after digging a few blocks
Environment	Operating System	Windows 11
	Browser	Chrome
Bug Details	Steps to reproduce	1. Launch the application. 2. Launch level3 3. Dig 6+ rows down 4. Open menu

	Expected results	Menu opens up in a clean rectangle in the middle of the screen
	Actual results	The menu may be visible on screen or not, and its size and content vary.
Bug Tracking	Severity	high
	Priority	Medium
Notes	Notes	Seems to become worse the lower you dig

Category	Label	Value
Bug ID	ID Number	BUG-005
	Name	Level3 - Star saving
	Reporter	Arol Nokam
	Submit Date	March 19, 2026
Bug Overview	Summary	The stars are saved on loss
Environment	Operating System	Windows 11
	Browser	Chrome
Bug Details	Steps to reproduce	1. Launch level3 with 0 stars previously saved 2. Collect 1 or 2 stars. 3. Lose (no more oxygen)
	Expected results	When returning to the level select screen, it should still display 0 stars
	Actual results	The level select screen displays the number of stars collected during the lost game
Bug Tracking	Severity	medium
	Priority	Medium

Notes	Notes	
-------	-------	--

Category	Label	Value
Bug ID	ID Number	BUG-006
	Name	Level2 - Esc
	Reporter	Arol Nokam
	Submit Date	March 19, 2026
Bug Overview	Summary	Level is exited on Esc
Environment	Operating System	Windows 11
	Browser	Chrome
Bug Details	Steps to reproduce	1. Launch level 2 2. Press Esc
	Expected results	Pause menu opens
	Actual results	Level select screen opens
Bug Tracking	Severity	high
	Priority	Medium
Notes	Notes	Not sure how it affects star tracking, more testing needed.+

Category	Label	Value
Bug ID	ID Number	BUG-007
	Name	Level 2 asset error
	Reporter	Gauthier Appaix

	Submit Date	March 24, 2026
Bug Overview	Summary	Asset for hook showing up as black box and console showing failure to load asset.
Environment	Operating System	Windows 11
	Browser	Brave, Chrome
Bug Details	Steps to reproduce	3. Launch level 2 4. Observe hook
	Expected results	Hook asset exists
	Actual results	Hook asset is replaced with black box
Bug Tracking	Severity	high
	Priority	high
Notes	Notes	Only asset with this problem detected in testing.

Resolution

Bug. ID	Status	Fixer Notes
BUG-001	Resolved ▾	Fixed By - Ebitimi Dambo Music was not stopped on LoginScene.js shutdown. I have added this feature, and pushed it to the repository.
BUG-002	Resolved ▾	Fixed By - Ebitimi Dambo The logout button's display property was set to none during the scene's shutdown() method, but it was not explicitly restored in the create() method when the scene was re-entered.
BUG-003	Resolved ▾	Fixed By - Ebitimi Dambo This bug was tricky, but I just set the loop variable to false, to tell it more explicitly that it should not repeat.
BUG-004	Resolved ▾	Fixed by - Arol Nokam The menu was set to a fixed position and was scrolled past by the camera as the player digs deeper. By setting the scrollFactor of the pause menu and its element to 0, it was allowed to properly be on-screen at all times.
BUG-005	Resolved ▾	Fixed by - Arol Nokam The level was missing a flag to decipher whether the exit of the level was done in a winning or a losing scenario. Adding the flag and integrating it to the SaveLevelProgress() allowed for a consistent save on win, and none on loss.
BUG-006	Resolved ▾	Fixed By - Aadil Bashir Pressing Esc opens the pause menu and pressing Y afterwards redirects to the title card. From there you can press to go in the level, or esc to navigate to the level select screen.
BUG-007	Resolved ▾	Fixed By - Aadil Bashir Deployment using github is case sensitive. So, while it worked in Phaser.js, I had to change the name of the hook path in the preload method to make it exact.